Experiment: Sqoop Installation and Execution of Basic Commands

Aim

To install and configure Apache Sqoop and execute basic commands to transfer data between relational databases and Hadoop.

Theory

Apache Sqoop is an open-source tool used to transfer data between relational databases (RDBMS) and the Hadoop ecosystem. It helps in importing data from databases such as MySQL, Oracle, and SQL Server into Hadoop Distributed File System (HDFS), Hive, and HBase. It also supports exporting processed data from Hadoop back into relational databases.

Sqoop stands for SQL-to-Hadoop. It is mainly used when organizations need to transfer large amounts of structured data between traditional databases and Hadoop for big data processing.

Features of Sqoop

1. Data Import: Imports data from relational databases into HDFS, Hive, and HBase.
2. Data Export: Exports data from Hadoop to relational databases.
3. Parallel Processing: Divides data into multiple parts and transfers them simultaneously.
4. Efficient Data Transfer: Supports transferring large datasets efficiently.
5. Integration: Works with Hadoop, Hive, and HBase.
6. Incremental Import: Imports only new or updated records instead of transferring the entire dataset again.

Installation and Configuration of Sqoop

The basic steps involved in installing and configuring Sqoop are:

1. Install and configure Java (JDK) on the system.
2. Install and configure Hadoop.
3. Download the Apache Sqoop package.
4. Extract the downloaded package into the required directory.

5. Configure environment variables such as SQOOP_HOME and update the system PATH.
6. Add the required JDBC driver, such as the MySQL Connector/J driver, to the Sqoop library directory.
7. Verify the installation using the following command:

sqoop version

Basic Sqoop Commands

Sqoop provides different commands to import, export, and list database information.

1. List Databases

Displays all databases available in the MySQL server.

```
sqoop list-databases \
--connect jdbc:mysql://localhost:3306/ \
--username root --password password
```

2. List Tables

Displays all tables present in a particular database.

```
sqoop list-tables \
--connect jdbc:mysql://localhost:3306/student_db \
--username root --password password
```

3. Import Data

Imports a table from MySQL into HDFS.

```
sqoop import \
--connect jdbc:mysql://localhost:3306/student_db \
--username root --password password \
--table students \
--target-dir /user/hadoop/students
```

4. Import Data into Hive

Imports a database table directly into a Hive table.

```
sqoop import \
--connect jdbc:mysql://localhost:3306/student_db \
--username root --password password \
--table students \
--hive-import
```

5. Export Data

Exports data from HDFS into a MySQL table.

```
sqoop export \
--connect jdbc:mysql://localhost:3306/student_db \
--username root --password password \
--table students \
--export-dir /user/hadoop/students
```

Advantages of Sqoop

- Simplifies data transfer between RDBMS and Hadoop.
- Supports large datasets through parallel processing.
- Reduces manual effort in data migration.
- Supports integration with Hive and HBase.
- Provides commands for both importing and exporting data.

Conclusion

Apache Sqoop is a useful tool for transferring structured data between relational databases and the Hadoop ecosystem. Its basic commands help perform database listing, table listing, data importing, and exporting. It plays an important role in big data environments by connecting traditional databases with Hadoop.

Note: Apache Sqoop is a legacy project and is not included in newer Hadoop distributions by default. Installation may require compatible Java, Hadoop, and JDBC driver versions.